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# XYZFlow: Scaling Multidimensional Shortcut Flows for Efficient Generative Modeling

*Adding Temporal History and Spatial Context to Flow Matching Straightens Trajectories, Enabling High-Quality Generation in Fewer Steps Without a Teacher Model*

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Diffusion models produce state-of-the-art images but require dozens of iterative sampling steps. The standard acceleration path — distilling a slow teacher into a few-step student — inherits the teacher’s quality ceiling and demands careful training. XYZFlow takes a different route: it redesigns flow matching itself by conditioning probability paths on denoising history (temporal dimension Y) and neighboring spatial patches (spatial dimension Z), in addition to the standard denoising time (dimension X). Richer conditioning straightens the flow trajectory, enabling high-quality generation in far fewer steps with no teacher model required.

## AT A GLANCE

**Problem:** Flow/diffusion models require many sampling steps for high quality; distillation-based speedups depend on teacher quality and are hard to train.

**Why it matters:** Efficient generation is critical for interactive and high-throughput image synthesis applications.

**Method:** Extend flow matching to three conditioning dimensions: denoising time (X), denoising history (Y), and spatial patch context (Z).

**Result:** Higher quality per function evaluation than distilled few-step models; no teacher model required; quality improves monotonically with conditioning.

**Evidence:** FID and CLIP-score improvements over 1–4 step baselines across multiple image benchmarks.

## The Big Picture

Flow matching learns a continuous-time transformation that maps noise  $z \sim \mathcal{N}(0, I)$  to data  $x \sim p_{\text{data}}$  via a probability path  $(p_t)_{t \in [0,1]}$ . Standard flow matching parameterizes this path along a single temporal dimension  $t$

(the denoising time), yielding a velocity field  $v_\theta(z, t)$  that must integrate to transform noise into data in a finite number of steps.

The core challenge is trajectory curvature: curved probability paths require many small integration steps for accurate traversal. Straight paths can be traversed in fewer steps with larger step sizes. Existing efficient methods reduce curvature after the fact via distillation, but this is computationally expensive and inherits the teacher model’s limitations.

XYZFlow introduces two additional conditioning dimensions that straighten the path *by design* during training: the denoising history (accumulated context from prior steps) and the spatial context (generated patches in neighboring regions). By conditioning on more of the world state at each step, the model’s uncertainty about the optimal next step decreases, and the induced trajectory becomes straighter.

## Why Existing Approaches Fall Short

**One/two-step distillation** (Consistency Models, LCM, TurboFlow) trains a student model to match the teacher’s output in one or two steps. Quality is bounded by the teacher, and the distillation objective is sensitive to hyperparameters. The student also cannot generalize beyond the teacher’s mode coverage.

**Adversarial diffusion distillation** (ADD, DMD) adds a discriminator to the distillation loss, improving one-step quality but further complicating training and introducing GAN instabilities.

**Autoregressive generation** achieves efficiency through sequential patch prediction but sacrifices the parallelism of diffusion-based approaches and requires discrete tokenization.

XYZFlow achieves the efficiency gains of distilled models without a teacher, and the expressivity gains of autoregressive models without discrete tokenization.

## Core Mechanism

XYZFlow’s velocity field is conditioned on three quantities at each step:

- **X — denoising time  $t$ :** Standard conditioning on current noise level, shared with all flow matching models.
- **Y — denoising history  $h_t = (z_{t_1}, z_{t_2}, \dots)$ :** A memory of the partial trajectory, conditioning on states at a set of prior denoising times  $t_1 > t_2 > \dots > t$ . This non-Markovian conditioning allows the model to exploit accumulated context about the direction of the trajectory.
- **Z — spatial context  $s_i$ :** During spatial dimension

generation (sequential patch prediction), conditioning on the set of already-generated patches  $s_{<i}$  provides spatial consistency that reduces uncertainty about each new patch’s target state.

The velocity field is:

$$v_\theta(z_t, t, h_t, s_{<i}) \approx \nabla_{z_t} \log p_t(z_t \mid h_t, s_{<i})$$

with richer conditioning providing a more informative gradient direction at each step.

### Technical Formulation

Standard flow matching minimizes the conditional flow matching loss:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, z_0, z_1} [\|v_\theta(z_t, t) - (z_1 - z_0)\|^2]$$

where  $z_t = (1 - t)z_0 + tz_1$  is the linear interpolation path. XYZFlow extends this to the multidimensional conditioned loss:

$$\mathcal{L}_{\text{XYZ}}(\theta) = \mathbb{E}_{t, z_0, z_1, h_t, s_{<i}} [\|v_\theta(z_t, t, h_t, s_{<i}) - (z_1 - z_0)\|^2]$$

The key theoretical result is that the *optimal* velocity  $v^*(z_t, t, h_t, s_{<i})$  is straighter (lower curvature) than  $v^*(z_t, t)$  whenever the conditioning variables  $h_t$  and  $s_{<i}$  are informative about the target  $z_1$ . This follows from the variance decomposition:

$$\mathbb{E}[\|z_1 - z_0\|^2] \geq \mathbb{E}[\|z_1 - \mathbb{E}[z_1 \mid h_t, s_{<i}, z_t]\|^2]$$

The residual uncertainty under conditioning is lower, implying the conditioned target is closer to the true target at each step.

### Learning or Inference Procedure

Training follows standard flow matching: sample pairs  $(z_0, z_1)$ , sample time  $t$ , compute  $z_t$ , retrieve history  $h_t$  from prior steps of the current trajectory, retrieve spatial context  $s_{<i}$  from previously generated patches, and minimize  $\mathcal{L}_{\text{XYZ}}$ .

At inference, generation proceeds in two nested loops:

1. **Temporal loop:** Take  $N_T$  denoising steps ( $N_T = 2-8$  for efficient generation, vs. 50–1000 for standard diffusion). Maintain history  $h_t$  across steps.
2. **Spatial loop:** At each denoising step, generate patches sequentially, conditioning each patch on the already-generated patches.  $N_S$  patches are generated per step.

### What the Guarantee Says

The paper proves that, under perfect conditioning, the XYZFlow velocity field induces a straighter trajectory

than standard flow matching, requiring fewer integration steps for the same approximation error. The practical guarantee is: for a fixed number of function evaluations (NFE), XYZFlow achieves lower FID than unconditioned flow matching and distilled models at matched NFE.

### Experimental Findings

- **Quality vs. NFE:** At NFE= 2, XYZFlow outperforms LCM and TurboFlow by 3–5 FID points on ImageNet-256 and COCO.
- **No teacher required:** XYZFlow trained from scratch matches or exceeds the quality of distilled models trained from a strong teacher.
- **Monotonic improvement:** Adding Y conditioning over X improves FID by 2–3 points; adding Z conditioning further improves by 1–2 points.  $X < XY < XYZ$  holds consistently across NFE settings.
- **Scaling:** Quality scales gracefully with NFE; there is no saturation plateau seen in distilled models at 2–4 steps.

### Ablations and Interpretation

**History length:** Conditioning on 2–3 prior states (vs. just  $t - 1$ ) captures most of the benefit; conditioning on the full history provides marginal additional gain at higher memory cost.

**Spatial patch order:** Raster-scan order outperforms random order, confirming that spatial locality in the conditioning signal matters.

**Architecture:** Cross-attention over history states outperforms concatenation for integrating the Y and Z conditioning signals.

### Reference

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